

Implementing of a grid-connected PV energy system in building with medium consumption: A techno- economic case study

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ABSTRACT

Renewable energy sources facilitate decentralized electricity production, reducing dependence on fossil fuels and contributing to a sustainable energy era. Algeria is one of the countries implementing a policy aimed at energy transition, with buildings playing a major role in this process. They consume more energy than any other sector and therefore significantly contribute to climate change. While various projects related to energy efficiency are being carried out, they are mostly limited to rural and Saharan areas or low-consumption residential buildings. The objective of this paper is to demonstrate the feasibility of integrating a grid-connected photovoltaic system into a medium-consumption building located in one of Algeria's major cities, Constantine, to meet the building's electrical energy demand. This paper presents a methodology for optimally sizing a grid-connected photovoltaic system to supply power to the Institute of Nutrition, Food, and Agro-Food Technologies (INATAA) in Constantine, Algeria. The technical and economic aspects of the system's components were considered to determine the most feasible configuration, using real data from the institute and the HOMER Pro simulation software. The simulation results show that the grid-connected PV system, without storage, can meet the institute's energy demand (with 671,061 kWh/year of energy produced, of which 81.3 % is photovoltaic energy). This configuration offers the lowest costs, with a Net Present Cost (NPC) of \$748,413 and a Cost of Energy (COE) of \$0.0894/kWh. Therefore, integration of a grid-connected photovoltaic system in medium-consumption buildings can effectively meet energy needs while reducing costs. Thus, this study demonstrates the technical and economic feasibility of such system to support the energy transition in urban areas in Algeria.

1. Introduction

In recent years, environmental protection and sustainable development have become of increasing importance. These concepts arose from reflections on the detrimental effects of pollution caused by conventional energy sources. As a result, the global need to adopt new energy sources has become essential to reducing greenhouse gas emissions and limiting the rise in temperatures to 2 °C by 2100 [1]. In response, many countries have committed to strengthening their energy transition policies, aiming to integrate clean and renewable energy sources into a significant part of their energy mix. This global energy shift is particularly important for countries like Algeria, where the development of the energy sector is vital for ensuring social, economic, and industrial growth. With electricity demand growing at rates exceeding 5 % over the past decade [2], the Algerian government has introduced new directives for the development of renewable energies. These initiatives aim to reduce the share of hydrocarbons in the national energy balance,

positioning Algeria to better meet its growing energy needs while promoting sustainability.

To this end, Algeria has been committed to an energy transition policy for several years, aiming not only to secure its energy supply but also to meet its contractual obligations in reducing greenhouse gas emissions [3]. This policy includes the introduction of energy efficiency measures in the three sectors of transport, industry, and building [4]. The building sector, in particular, presents significant opportunities for research aimed at reducing energy demand. It is a non-productive but energy-intensive sector, consuming 41 % of final energy, ahead of the agricultural sector, which absorbs 33 %, as well as the industrial and transport sectors, with respective rates of 19 % and 7 %. Several factors accelerate domestic energy demand in this sector, notably the ever-growing housing stock, services (hotels, hospitals, etc.), and population growth. The goal is to reduce residential and tertiary energy consumption by 10 to 15 % by 2030 [2].

To achieve a successful and efficient energy transition, the government must fully harness Algeria's renewable energy. Algeria possesses

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Nomenclature			
INATAA	Institute of Nutrition, Food, and Agro-Food Technologies	NOCT	the nominal operating cell temperature [°C]
NPC	Net Present Cost [\$]	P _{in}	the inverter power [W]
COE	Cost of Energy [\$/KWh]	η _{lin}	the inverter efficiency [%]
PV	Photovoltaic	P _{ch}	the hourly power demand [W]
AC	Alternating current	SOC	battery state of charge [%]
DC	direct current	SOC _{max}	SOC upper limit [%]
KWh	kilowatt-hour	SOC _{min}	SOC lower limit [%]
KW	kilowatt	C _B	availability power of battery [Ah]
KWp	kilowatt-peak	σ	the self-discharge rate of battery bank
P _{PV}	PV power output [KW]	C _{Bmax}	allowable energy level remained in battery [Ah]
f _{PV}	PV derating factor	C _{Bmin}	allowable energy level remained in battery [Ah]
Y _{PV}	the rated capacity of the PV array [KW]	N _{bat}	total number of batteries
G _T	the global solar radiation [kW/m ²]	DOD _{max}	the maximum depth of discharge [%]
G _{T,STC}	irradiation at standard test conditions [W/m ²]	P _{load}	the load demand [W]
α _p	the temperature coefficient [%/°C]	η _{bat}	the efficiency of storage batteries [%]
T _c	PV cell temperature [°C]	C _{ACC}	the annualized capital cost [\$]
T _{c,STC}	the PV cell temperature under standard conditions [°C]	C _{ARC}	annualized replacement cost [\$]
T _a	the ambient temperature [°C]	C _{AOM}	annualized operation and maintenance cost [\$]
τ	the transmittance of the cover over PV array	E _{AEC}	the annualized electricity supplied [KWh]
α	the solar absorptance of the PV array	CRF	capital recovery factor
U _L	the coefficient of heat transfer to the surrounding [W/m ² K]	n	number of years
η _c	the PV array electrical conversion efficiency [%]	i	the real annual interest rate
		i'	the nominal interest rate and f is the annual inflation rate

one of the most significant renewable energy potentials globally, due to its vast territory, location, and favorable geographical conditions. Among the various types of energy sources, solar energy stands out as the most abundant and competitive, offering significant opportunities for future development in various forms. Algeria is one of the world's largest sources of solar energy, with an insolation period of 2000 to 6000 Wh/m², equivalent to 10 times the world's energy consumption [5]. However, despite this immense potential, the country is currently underutilizing renewable energy, with solar photovoltaic power accounting for only a small fraction of the energy mix. For instance, in 2020, 96 % of the electricity supply was generated by natural gas, while most renewable energies accounted for only around 4 %, with solar energy accounting for 54 % of the total electricity generated from renewable sources [6,7].

Despite the current low level of renewable energy utilization, the Algerian government is committed to combating climate change and improving the country's energy supply by launching a national renewable energy and energy efficiency program. Solar PV is playing a key role in this energy transition, with a capacity of 13.5GW [8], and the country aims to generate 27 % of its electricity from renewable sources, mainly solar, by 2035.

Solar energy, particularly the photovoltaic sector, represents a strategic lever for sustainable development in Algeria. It has the potential to play a key role in the country's energy transition, especially in the building sector, by reducing dependence on fossil fuels and lowering greenhouse gas emissions. However, despite the adoption of a national strategy in 2011 to promote renewable energies, the implementation of solar projects remains limited. These projects are primarily concentrated in the Saharan regions, the high plateaus, and isolated areas.

In this context, we are going to prove that the reliability and energy efficiency of PV systems are not limited to installations in rural and Saharan areas, but also in large cities with large populations and high energy (electricity) consumption, such as the city of Constantine. In this article, we will be sizing and optimizing a photovoltaic system connected to the grid (on-grid) to supply the INATAA institute in Constantine, Algeria. A techno-economic analysis will be carried out using the HOMER Pro (Hybrid Optimization Model for Electric Renewable) simulation software. HOMER Pro is a platform developed by the

National Renewable Energy Laboratory (NREL) to design, optimize and analyse hybrid systems meeting the desired energy requirements [9,10]. The software runs thousands of optimization iterations for each scenario and input considered [11]. In the end, the system with the lowest net present cost and the lowest cost of energy will be selected and proposed as the optimal solution for powering the INATAA institute.

2. Literature review

2.1. Overview of previous studies

A number of research studies have focused on the sizing and techno-economic analysis of grid-connected and off-grid photovoltaic systems. In this section we will highlight some of these research works.

Brahim et al. [12] proposed a grid-connected PV system for Abdelmalek Essadi University in Morocco. The results show that a grid-connected PV system with storage is the most economically feasible system.

Liaquat et al. [13] designed a hybrid diesel/PV/wind turbine system for the town of Mount Magnat in Australia. HOMER Pro software was used for the simulation. They found that a hybrid system consisting of a 400KW diesel generator, 500KW PV panels, 4 x 275KW turbines and 300 batteries with a 300KW inverter could save AUD 26.3 per KWh over the lifetime of the project.

Recently, using a similar approach, Shahid et al. [14] presented an economic analysis of the use of a PV/diesel/battery hybrid system to meet the electrical load of a typical residential building (with an annual electrical energy demand of 35,120 KWh) in five different areas of Saudi Arabia. The simulation results show that for a system consisting of 4Wp PV panels with a 4KW diesel generator and a storage system (battery) with 3 h autonomy, the PV system penetration rate is 22 %, 21 %, 22 %, 20 % and 20 % and the energy production cost (COE in US\$/KWh) is 0.179, 0.179, 0.178, 0.180 and 0.181 in Abha (southern zone), Hofuf (eastern zone), Qurayat (northern zone), Taif (western zone) and Riyadh (central zone) respectively.

Harpreet and Gagandeep [15] carried out the optimization and sensitivity analysis of a smart microgrid. The software used for this analysis was also HOMER and the results, after rigorous calculations,

found that the implementation of a smart microgrid at Gujral Punjab University in India would be feasible and cost effective. The study shows that installing solar PV panels in line with the government’s rooftop policy is economically beneficial and will ensure a clean and sustainable future.

Serdouk F. [16] carried out the modeling and simulation as well as the technical and economic potential of a small grid-connected photovoltaic power plant with a capacity of 1 MW in the town of L’Oued, Algeria. The system was simulated using SAM (System Advisor Model) software.

Majid et al. [17] proposed a design and optimization of a PV system using HOMER software computer model to supply lighting load for renewable energy lab in Sohar University in Oman. All the optimization systems are ranked according to net present cost, and all other economic outputs are calculated for the purpose of powering the lab and finding the best net present cost. The result obtained from the optimization gives the initial capital cost as 13.500 US\$ while operating cost is 817 US \$/year. Total net present cost is 23.939 US\$ and the cost of energy is 1.354 US\$/kWh.

Jedou et al. [18] presented a map of electricity consumption potential according to the electricity demand and to encourage renewable energy in terms of its integration into electrical energy generation systems (on-grid).

Das et al. [19] analyzed the economic and environmental benefits of both stand-alone and grid-integrated hybrid energy systems composed of PV/Wind/Diesel/Battery configurations in five different climatic regions in Bangladesh using the HOMER optimization model. A detailed techno-economic study of optimized hybrid systems is further examined by integrating the grid-connected option. Results show that the cost of energy and net present cost of the stand-alone hybrid PV/Diesel/Battery system are slightly lower in the Rajshahi region, while the Chattogram region demonstrates higher financial and environmental potential. The grid-connected hybrid energy system offers significant cost savings (\$0.07/kWh) compared to the grid tariff (\$0.10/kWh), with substantial excess energy supplied to the grid. Additionally, grid-connected systems could save around 45,582 kg-CO₂/year, compared to 32,905 kg-CO₂/year for stand-alone hybrid systems.

Anwari and Ayong in [20] studied the technical feasibility of off grid PV that generated 2.5 % of load requirement of Makkah city in Saudi Arabia based on solar irradiance using HOMER software. The load profile with random variability factors was assumed based on the load pattern in Makkah with no grid connection consideration.

In [21], they applied grid-connected solar PV to the same case. Similarly, Adaramola examined the feasibility of grid-connected solar PV in Jos, Nigeria, investigating the technical and economic performance of the system. The load profile was assumed based on the pattern of the energy consumption in Jos. He concluded that the solar PV system could provide for around 40 % of the annual electricity consumption, whereas, aside from the amount of solar irradiation, the initial cost of the scheme plays a significant role in electricity price.

2.2. Comparison and research contribution

This article presents a feasibility analysis and an evaluation of the technical and economic reliability of a grid-connected photovoltaic (PV) system, designed to meet the electricity demand of a medium-consumption building. While several similar studies on the integration of renewable energy in buildings have been conducted in Algeria, this work stands out due to several specific advantages, which are summarized as follows:

- The study is based on real data, whether technical, economic, or related to the selected building (including its consumption), ensuring an optimal sizing system.

Table 1
Solar energy projects carried out under the supervision of the ministry of the interior, local authorities, and territorial planning [7].

Applications	Accumulated capacities in 2019(KWp)
– Public lighting (404,551 new light points, including 15,150 powered by solar PV energy) – Additionally, 265,758 energy-consuming lamps in the existing public lighting network were replaced with LED luminaires.	1515
Electrification of 348 primary schools using solar PV installations, including 96 located in areas remote from the public electricity grid.	3610
Installation of 3,550 individual solar kits for households without any means of electrification.	4021
– Installation of 3,500 solar-powered light points along beaches and recreational forests. – Installation of solar charging stations for low-capacity battery recharging.	4197
Total	9146



Fig. 1. Location of Constantine city.

- The chosen building is of particular interest due to its location in a large city, the amount of electricity consumed, and its nature as a university institute.
- Unlike many previous studies that propose hybrid systems (wind/PV/battery) for isolated and Saharan sites, which are characterized by very high costs.

In addition, numerous projects have been implemented in Algeria under the national program for the promotion of renewable energy and the energy transition. Statistics indicate that the majority of solar installations are located in isolated or Saharan areas. Table 1 lists the solar energy projects carried out under the supervision of the Ministry of the Interior, Local Authorities, and Territorial Planning. These projects are primarily focused on public lighting applications, the electrification of low-consumption buildings (such as schools), and installations in remote areas. These data highlight the need to extend the use of photovoltaic technologies to other sectors to ensure a more effective and sustainable energy transition in Algeria. This study aims to encourage the Algerian government to increase investments in the integration of PV systems in

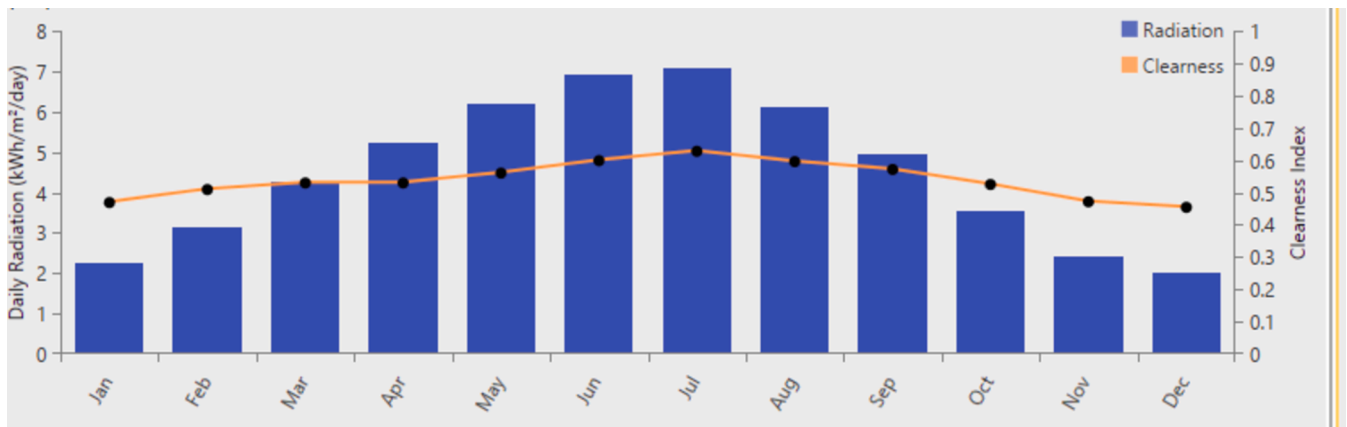


Fig. 2. Monthly profile of global solar radiation and clearness index of Constantine city.

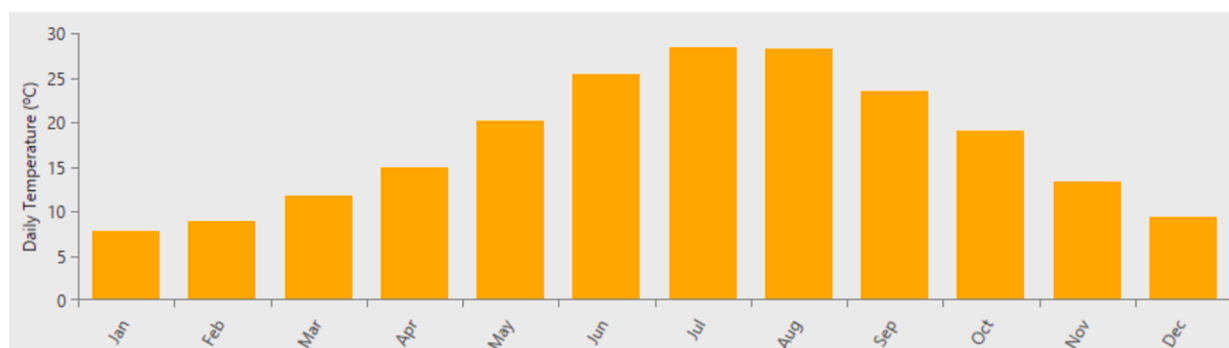


Fig. 3. Monthly average values of air temperature of Constantine city.

the building sector, particularly in large cities.

3. Methodology

3.1. Meteorological data

To design an optimal photovoltaic (PV) system, a thorough analysis of the chosen site is crucial. The site specifications are detailed in Fig. 1. The INATAA institute, located in eastern Algeria, experiences a

Mediterranean climate, with latitude of $36^{\circ} 24.2' N$ and a longitude of $6^{\circ} 37.9' E$. In this study, synthetic hourly solar data generated by HOMER from monthly average values were used, sourced from NASA's Surface Meteorology and Solar Energy database over a 22-year period (July 1983–June 2005).

The energy output of a PV system is influenced by solar irradiation and the ambient temperature of the PV array. HOMER incorporates these factors by utilizing monthly average global horizontal irradiance and temperature as key input parameters, as defined in its resources.

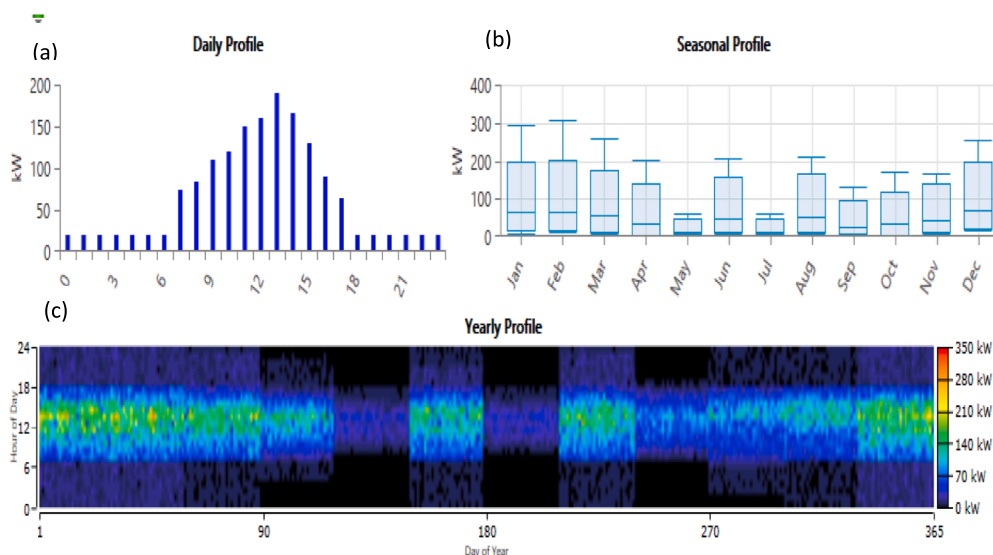


Fig. 4. INATAA load profile: (a) daily, (b) monthly, (c) annual.

Fig. 2 shows the clearness index and average monthly global solar radiation for the city of Constantine, with solar radiation peaking in June, July, and August, reaching an average of 6.7 kWh/m² per day.

The monthly average air temperature is depicted in Fig. 3. The annual average temperature is 17.6 °C, with a long summer season, from June to September, characterized by higher temperatures. This ambient temperature profile is important for determining PV system efficiency, as HOMER calculates the power output of the PV array by considering the cell temperature at each time step. It is important to note that PV system performance is influenced by various factors, including the PV cell temperature and the level of solar irradiation.

3.2. Load profile

Determining the load profile is the most important task for optimal sizing of any type of renewable energy system, as an inappropriate load profile will lead to under sizing or oversizing with negative consequences, especially in terms of system cost and management.

Fig. 4 shows the monthly, daily and yearly power consumption profile of the INATAA Institute. The institute has an annual energy consumption of approximately 378,511 kWh, with a reduced average daily energy consumption of 1045.31 kWh/day. This consumption is expected to increase with the growth in student numbers and laboratory equipment.

The highest demand for charging occurs annually from November to February with an average of 200 kW, whilst the month with the least demand for charging is July with a value of approximately 13 kW. During working hours, from 8 am to 5 pm, the institution's charging profile displays a large charging consumption. Subsequently, this trend progressively decreases between 5 pm and 8 am.

3.3. Mathematical modeling and economic criteria

3.3.1. Modeling of the different system components

a) PV panel

HOMER models the PV array as a device that produces DC electricity in direct proportion to the global solar radiation incident upon it, independent of its temperature and the voltage to which it is exposed. The power output of the PV array is calculated using the following equation [22]:

$$P_{PV} = f_{PV} \cdot Y_{PV} \cdot \left(\frac{G_T}{G_{T,STC}} \right) \cdot [1 + (T_c - T_{c,STC})] \quad (1)$$

Where, f_{PV} is the PV derating factor, Y_{PV} the rated capacity of the PV array (kW), G_T the global solar radiation (beam plus diffuse) incident on the surface of the PV array (kW/m²), and $G_{T,STC}$ (W/m²) is irradiation at standard test conditions (i.e. 1000 W/m²), α_p (%/°C) is the temperature coefficient, T_c (°C) is the PV cell temperature, and $T_{c,STC}$ (°C) is the PV cell temperature under standard conditions. The cell temperature can be calculated using the energy balance for the PV array as follows:

$$T_c = T_a + G_T \left(\frac{\tau \alpha}{U_L} \right) \left(1 - \frac{\eta_c}{\tau \alpha} \right) \quad (2)$$

Where T_a (°C) is the ambient temperature, τ is the transmittance of the cover over PV array, α is the solar absorptance of the PV array, U_L (W/m²K) is the coefficient of heat transfer to the surrounding, and η_c is the electrical conversion efficiency of the PV array. The cell temperature under the condition of $G_T = 800$ W/m², $T_a = 20$ °C, and $\eta_c = 0$ (no load operation) is called nominal operating cell temperature (NOCT) and is reported by manufacturers.

$$\frac{\tau \alpha}{U_L} = \frac{T_{c,NOCT} - T_{a,NOCT}}{G_{T,NOCT}} \quad (3)$$

and

$$T_c = T_a + G_T \left(\frac{T_{c,NOCT} - T_{a,NOCT}}{G_{T,NOCT}} \right) \left(1 - \frac{\eta_c}{\tau \alpha} \right) \quad (4)$$

b) The inverter

The power P_{in} (in W) supplied by the inverter to meet the demand is given by [23]:

$$P_{in} = \frac{P_{ch}}{\eta_{in}} \quad (5)$$

When η_{in} is the inverter efficiency specified by the manufacturer (%) and P_{ch} is the hourly demand (W).

c) Battery bank

The battery output current depends on the output power of the renewable generators, the AC load, the control strategy and its SOC [24]. The battery charging and discharging control with the maximum power of PV array is the key point to increase the efficiency of the generation system [25]. Energy production and absorption of battery system during the time from $t-1$ to t is described by the equation below:

$$C_B = C_B(t-1)(1 - \sigma) + \left(P_{PV}(t) - \frac{P_{load}(t)}{\eta_{inv}} \right) \eta_{bat} \quad (6)$$

When

$P_{PV}(t) - \frac{P_{load}(t)}{\eta_{inv}} > 0$ and $C_B(t-1) < C_{Bmax}$ the battery is charging.

$P_{PV}(t) - \frac{P_{load}(t)}{\eta_{inv}} < 0$ and $C_B(t-1) > C_{Bmin}$ the battery is discharging.

Where $C_B(t)$ is the availability power of battery bank at hour t and $C_B(t-1)$ is the availability power of battery bank at the previous hour ($t-1$). The term σ is the self-discharge rate of battery bank. $P_{load}(t)$ is the load demand at time step of t . η_{bat} is the efficiency of storage batteries and η_{inv} is the efficiency of the inverter. The value of $C_B(t)$ is between the minimum allowable energy level remained in battery banks (C_{Bmin}), and the maximum allowable energy level (C_{Bmax}):

$$C_{Bmin} \leq C_B(t) \leq C_{Bmax}$$

For longevity of the battery bank, the maximum charging rate (SOC_{min}) is given as the upper limit, and it takes the value of total nominal capacity in Ah of the battery bank which is defined by the total number of batteries (N_{bat}), and the nominal capacity of each battery (C_B), as follows [26]:

$$SOC_{max} = C_B N_{bat} \quad (7)$$

The minimum permissible state of charge (SOC_{min}) of battery bank during discharging may be expressed as follows:

$$SOC_{min} = N_{bat} C_B (1 - DOD_{max}) \quad (8)$$

Where DOD_{max} is the maximum depth of discharge in percentage. The battery can have the maximum life if the DOD is set between 30 and 50 %.

Temperature can also affect the performance of batteries. High temperature decreases the battery lifetime. The batteries are normally installed inside a building where the temperature is not expected to change drastically. It is recommended to keep the battery at 25 °C [27].

3.3.2. Economic criteria

In this study, the optimal design is determined based on the lowest NPC and COE of the system configurations. HOMER determines the COE using equation (9) [28], where C_{ACC} , C_{ARC} and C_{AOM} represent the annualized capital cost, annualized replacement cost and annualized operation and maintenance cost respectively, while E_{AEC} represents the annualized electricity supplied.

$$COE = \frac{C_{ACC} + C_{ARC} + C_{AOM}}{E_{AEC}} \quad (9)$$

On the other hand, the NPC is calculated from Equation (10), whereby $CRF(i, n)$ is the capital recovery factor and can be found by using

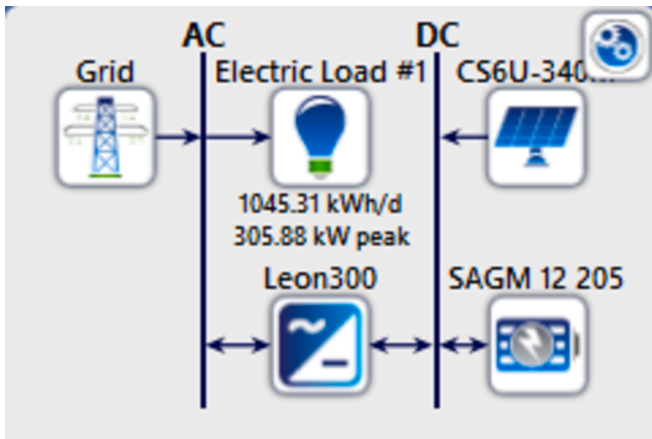


Fig. 5. Schematic of the proposed system.

Table 2
Details of system components.

PV array	
Technology	Canadian Solar Max Power CS6U340M
Peak power (W)	340
Efficiency (%)	17.49
Capital cost (\$/kW)	1600.10
Life time (year)	25
Nominal operating cell temperature (°C)	45
Inverter	
Technology	LeonicsMTP-4117
Power (KW)	300
Efficiency (%)	96
Capital cost (\$/KW)	300
Life time (year)	10
Battery	
Technology	Trojan SAGM 12 205
Nominal capacity (KWh)	2.63
Maximal capacity (Ah)	219
Voltage (V)	12
Capital cost (\$)	1700
Life time (year)	10

Equation (11) [29].

$$NPC = \frac{C_{ACC} + C_{ARC} + C_{AOM}}{CRF(i, n)} \quad (10)$$

$$CRF(i, n) = \frac{i(1 + i)^n}{(1 + i)^n - 1} \quad (11)$$

Where n is the number of years, and i is the real annual interest rate, which is calculated using equation (12), whereby, i' is the nominal interest rate and f is the annual inflation rate. The present study considers

a 10 % discount rate and a 2 % possible inflation rate.

$$i = \frac{i' - f}{1 + f} \quad (12)$$

3.4. Technical description of the PV system

The selection and sizing of the components of the proposed PV system is carried out using the HOMER pro software. The system considered in this study is a photovoltaic system connected to the national electricity distribution network (on-grid system). The diagram of the proposed system is shown in Fig. 5. The main components of this system are: photovoltaic panels, inverter of the Leonics GTP-507 type with a capacity of 125KW, and batteries of the Trojan SIND 06920 model. Table 2 provides more details on the technical and economic characteristics of the system.

3.5. Grid integration

In this study, electricity from the grid is used to meet the load demand when renewable energy sources are unable to fully satisfy it. The photovoltaic system operates only during the daytime. When the solar power generation exceeds the primary load demand, the surplus electricity is sold back to the grid. This strategy reduces the need for large storage capacity, as a significant portion of the excess energy, typically wasted in off-grid systems, is fed into the national grid. For the simulation design, a fixed unit energy purchase cost (COE) of \$0.16/kWh was selected for residential applications, while a fixed sell-back price of \$0.08/kWh was chosen for the electricity injected into the grid.

4. Results and discussions

The present research analyses the potential of a grid connected PV system (microgrid) for a medium-consumption building in the city of Constantine, Algeria. The configuration of each proposed system is determined based on the cost-effectiveness and environmental indicators. The results are evaluated in terms of NPC, COE, PV capacity, battery capacity, inverter capacity, grid contribution, electric production, consumption, excess energy generation, and PV fraction. The analysis of the various results obtained from the simulation is summarized in the following sections.

4.1. Techno- economic analysis of the proposed systems

The optimal design of a grid-connected solar PV system is determined through simulations using the HOMER Pro software. Meteorological, technical and economic parameters described in Sections 3, 4 and 5 are considered. The nominal capacity used includes the efficiency and surface area of the PV module to meet the electrical energy needs of the Institute. Excess energy is stored in batteries until maximum capacity is reached, with the batteries serving to import and export electrical energy during power shortages. An inverter facilitates the transfer of energy between the DC and AC components in this system. Table 3, along with Fig. 6; illustrate electrical production and consumption for

Table 3
Results of energy produced and consumed.

Case	Microgrid plan	Electric production (KWh)	Electric consumption (KWh)	Excess electricity (%)	PV fraction (%)
Case1		671,068	647,630	0.224	81.3
Case2		678,555	654,351	0.288	81.6
Case3		381,538	381,538	0	0
Case4		381,538	381,538	0	0



Fig. 6. (a) The power produced by the PV panels; (b) comparison of energy produced between the grid and the PV modules.

various configurations of the suggested system. In fact, the optimal configuration (Case 1: PV system connected to the grid without storage) produces 671,068 KWh/year of electrical energy compared to a consumption of 647,630 KWh/year, resulting in an excess of electrical energy of 0.224 %. In this case, the percentage of photovoltaic electricity in the system is 81.3 %. The second case involves a grid connected system with storage, producing 678,555 kWh/year and consuming 654,351 kWh/year. The third and fourth cases rely solely on the electricity grid, except that the last system includes batteries; these two configurations have the same amount of energy produced and consumed, totaling 381,538 kWh/year. From Table 2, we can observe that the power generated by the system proposed in the second case (grid-connected system with storage) is greater than that of the optimal system (case 1: micro grid without storage). This is attributed to the fact that the HOMER Pro software selects the optimal configuration based on the lowest NPC.

The energy produced by the photovoltaic system supplies the peak load demand. Fig. 6(a) shows the typical monthly evolution of photovoltaic solar energy production. It can be observed from the graph that the energy output from the PV panels reaches significant values, ranging between 320 kW and 400 kW, especially during solar noon.

Furthermore, this output increases during the months of May, June, July, and August, due to the higher solar irradiation intensity during this period. Fig. 6(b) compares the average energy production between the electrical grid and the photovoltaic panels. It can be seen that the total energy generated by the PV panels exceeds that of the grid, particularly in January and December, where the energy produced by the grid is almost the same as that of the PV panels, due to atmospheric variations such as cloud cover and humidity.

• Configurations proposed

Different configurations are proposed based on two economic criteria, namely the Net Present Cost and Cost Of Energy, on which the simulated system optimization is based. HOMER Pro software eliminates insignificant solutions and selects the configuration that is most suitable based on the minimum NPC.

The minimum NPC and COE are achieved through a combination of systems defined for each case. The economic indicators for the optimal micro grid plan are displayed in Table 4.

This table presents the various alternatives that were simulated and optimized using the HOMER Pro software. The electricity distribution

Table 4
Results of proposed configurations.

Case	Microgrid plan	PV (KW)	Battery	Grid (KW)	Inverter (KW)	NPC (\$)	COE (\$)
Case1		360	--	999,999	300	748,413	0.0894
Case2		365	3	999,999	300	753,654	0.0891
Case3		--	--	999,999	--	789,175	0.160
Case4		--	1	999,999	--	988,423	0.200

Table 5
CO₂ emissions for each system.

Cases	Emissions of CO ₂ (Kg/year)
Case 1	76,619
Case 2	76,086
Case 3	241,132
Case 4	241,132

network was selected as the base system to determine the optimal approach.

Case 1: which features a PV system connected to the grid without storage with 360 KW solar panels, is the optimal configuration. The NPC is at its lowest, costing 748,413\$, and the COE is 0.0894\$/KWh.

Case 2: This scenario is solely reliant on solar PV panels, with a capacity of 365 kW, battery storage, an inverter, and the national electricity grid. The NPC is 753,654\$, and the COE for this case is 0.0891 \$/KWh.

Case 3: This scenario showcases the conventional grid alone, which serves as the fundamental system of the simulation. The NPC is 789,175 \$, and the COE is 0.160\$/KWh.

Case 4: This scenario also includes the electricity distribution network, along with battery storage and a inverter. This configuration has the highest NPC at 988,423\$ and a COE of 0.200\$/KWh.

4.2. Environmental emissions

The annual environmental emissions are directly linked to the amount of fossil energy generated by electric power plants. Table 5 presents the quantity of CO₂ emissions in kg/year for the proposed configuration. It is important to note that in the last two cases, the share of renewable energy is zero, resulting in a higher contribution to greenhouse gas emissions, particularly CO₂. These findings confirm that conventional electricity production networks are a major contributor to the rise in greenhouse gas emissions. Specifically, the grid-connected system alone (cases 3 and 4) generates a substantial amount of CO₂, amounting to 241,132 kg/year. In contrast, grid-connected PV systems, with or without storage, exhibit significantly lower CO₂ emissions (case 1 with 76,619 kg/year and case 2 with 76,086 kg/year). Thus, it is clear that the grid-connected PV option not only offers financial benefits but is also environmentally sustainable.

4.3. Sensitivity analysis

A sensitivity analysis was conducted to evaluate the impact of varying input parameters on system performance and to identify the most influential factors. In this study, the parameters examined are load consumption and solar irradiation for the optimized grid-connected PV system. The results of this analysis indicate that the NPC remains comparable when the daily load demand varies from 500 kWh to 1200 kWh, while keeping solar irradiation constant, as shown in Fig. 7. For instance, the COE is found to range between \$0.160/kWh and \$0.093/kWh at a fixed solar irradiation of 4.51 kWh/m²/day. Moreover, the COE decreases from \$0.089/kWh to \$0.055/kWh as solar irradiation increases from 4.51 kWh/m²/day to 7.08 kWh/m²/day, at a load demand of 1060 kWh/day.

5. Conclusion

This paper investigates the design, sizing, and optimization of a grid-connected photovoltaic (PV) system for supplying electricity to the INATAA institute in Constantine, Algeria. The feasibility and reliability of the system were evaluated using HOMER Pro simulation software, alongside real electricity consumption data from the institute. The two main parameters used to identify the most economically feasible energy system were the NPC in (\$) and the COE, in (\$/kWh).

Four potential scenarios were proposed through the simulation. The analysis revealed that a grid-connected PV system without storage can sufficiently meet the institute's energy demand, producing 671,061 kWh/year, with 81.3 % of the energy derived from photovoltaic sources. This configuration demonstrated the lowest costs, with an NPC of \$748,413 and a COE of 0.0894\$/kWh. Compared to the grid-only system, this configuration reduces CO₂ emissions by approximately 164,513 kg/year, representing a 68 % reduction.

These results highlight the feasibility of implementing photovoltaic systems in large buildings, such as university campuses, to meet their electricity needs. The economic benefits of investing in grid-connected PV systems provide strong incentives for the Algerian government to promote the installation of these systems, contributing to the country's energy transition and advancing sustainability in renewable energy. The findings of this study serve as a useful tool for future research. It is recommended that further studies explore other renewable energy sources (e.g., concentrated solar power, wind energy, and biogas) and various control strategies in the design of energy systems for urban

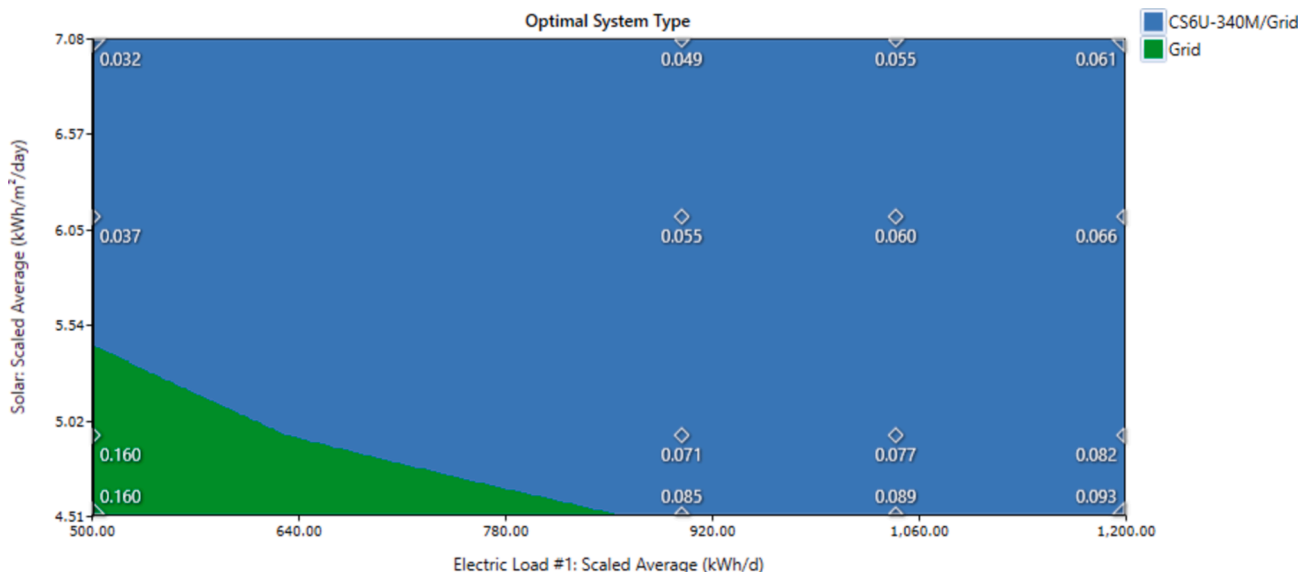


Fig. 7. Sensitivity of load demand and solar irradiation for PV connected system.

buildings.

CRediT authorship contribution statement

Louiza Bouseba: Writing – original draft, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Abla Chaker:** Writing – review & editing, Project administration, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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